Donda Dheeraj Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	7.58
Narayana Junior College	Class XII (Telangana State Board of Intermediate Education)	2024	98.20%
Mount Litera Zee High School	Class X (CBSE)	2022	96.40%

SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL.

Web Technologies: React, Next.js, Express.js, Node.js, Socket.IO, Prisma ORM, EJS, jQuery.

Databases & Caching: PostgreSQL, MySQL, SQLite, MongoDB, Redis.

Dev Tools: Git, GitHub, Docker, Postman.

Concepts: Qiskit, Data Structures & Algorithms, OOP, DBMS, API Design, Caching, Rate Limiting, Distributed Systems.

PROJECTS

Quantum Portfolio Optimization | Python, Qiskit, Qiskit Finance, NumPy, Matplotlib

- Formulated a 9-asset portfolio selection problem as a **QUBO** using Markowitz mean-variance optimization, balancing return against risk under a fixed budget constraint.
- Benchmarked 3 solvers -- an exact classical eigensolver, **VQE**, and **QAOA** -- comparing quantum heuristic solutions against the classical optimum across **512 candidate portfolios**.
- Implemented a parameterized quantum circuit (**ansatz**) with full entanglement and tuned it via **COBYLA** within a hybrid quantum-classical optimization loop.
- Derived return and covariance statistics from historical price data to construct the optimization objective.

URL Shortener | Next.js, Express.js, PostgreSQL, Redis, JWT

- Developed a multi-tenant URL shortener with **JWT authentication**, protected APIs, and user-specific link ownership.
- Added **Redis caching** for hot URLs, improving redirect throughput to **1,490+ RPS** with **66ms average latency** under **100 concurrent connections**.
- Implemented a **sliding-window rate limiter** to protect APIs from abuse while maintaining stable request handling under load.
- Designed a cache-first redirect flow with **24-hour TTL-based eviction**, achieving **100% successful HTTP 302 redirects** during sustained traffic spikes.

Distributed Job Queue System | Node.js, TypeScript, Redis, PostgreSQL

- Built a distributed job queue capable of processing **450+ jobs/sec** (3,000-job load test) with configurable worker concurrency and priority-based scheduling.
- Used **Redis Sorted Sets** for delayed job scheduling and reliable job assignment across workers.
- Added **retries with exponential backoff**, and **stalled-job recovery** to safely handle failed or abandoned jobs, ensuring no job is permanently lost due to worker crashes
- Built REST APIs for job submission, and queue monitoring with **API-key authentication** and full job history persistence in PostgreSQL.

Collaborative Project Management System | Next.js, TypeScript, PostgreSQL, Prisma, Socket.IO, Redis

- Built a multi-tenant project management platform with **10,000+ seeded tasks** to simulate production-scale workloads.
- Added **composite indexes** and asynchronous background writes, reducing critical-path query latency to **~3ms**.
- Improved drag-and-drop Kanban reordering from **O(n) to O(1)** using **fractional indexing**, reducing updates to a single-row write.
- Built real-time collaboration using **Socket.IO** for low-latency multi-client synchronization.

ACHIEVEMENTS

- Solved **1000+** Data Structures and Algorithms problems across leading competitive coding platforms, including Codeforces, AtCoder, LeetCode, and CodeChef.
- Achieved **3 Star (1614 rating)** on CodeChef.
- Secured **Global Rank 327** in CodeChef Starters Round 221.

RELEVANT COURSES

Math and Computer Science Courses: Introduction to Programming, Introduction to Probability, Statistics, Linear Algebra, Calculus, Complex Variables, Transform Techniques.

Physics: Engineering Mechanics, Fluid Mechanics, Hydraulic Engineering, Thermodynamics

EXTRACURRICULAR

- Won the Under-19 Inter-School Football Tournament, securing the **Gold Medal**.
- Won a Gold Medal in the 100 m Sprint and a Silver Medal in High Jump at the school athletics meet.
- Represented the school football team for **4 years**, competing in inter-school tournaments.